

## VERKHNEIMBATSK, Russian Federation

WMO: 236780

Lat: 63.1555N

Lon: 87.9667E

Elev: 149

StdP: 14.62

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -50.7      | -44.6 | -54.9                         | 0.2 | -50.9 | -49.2 | 0.3 | -44.8 | 14.3                  | -0.6 | 13.0 | 1.1  | 1.1                   | 100  | 0.680 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.1                   | 82.5            | 65.7 | 78.7 | 64.2 | 74.9 | 62.8 | 68.0                | 78.0 | 66.0 | 75.3 | 64.2 | 72.7 | 4.9                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.6                            | 91.8 | 71.8 | 62.6 | 85.2 | 69.8 | 60.6 | 79.5 | 68.8 | 32.4          | 78.2 | 30.9 | 75.4 | 29.5 | 72.9 | 81.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 13.9              | 12.0 | 10.6 | -53.7                      | 87.9 | 4.9                | 2.9 | -57.2                                              | 90.0 | -60.0      | 91.7 | -62.7      | 93.4 | -66.3      | 95.5 |
|                   |      |      | -52.6                      | 71.8 | 6.1                | 3.2 | -57.0                                              | 74.1 | -60.6      | 76.0 | -64.0      | 77.8 | -68.5      | 80.1 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 25.6   | -14.7 | -3.2  | 13.1  | 26.2  | 39.4  | 57.7  | 63.3  | 56.2  | 44.1  | 28.0  | 1.8   | -6.3  |
|                                                                               | DBStd       | 29.05  | 17.92 | 14.65 | 14.22 | 11.85 | 9.65  | 9.58  | 7.05  | 6.72  | 7.77  | 10.96 | 17.71 | 18.54 |
|                                                                               | HDD50       | 9849   | 2007  | 1489  | 1143  | 713   | 355   | 37    | 1     | 17    | 212   | 684   | 1445  | 1746  |
|                                                                               | HDD65       | 14488  | 2472  | 1909  | 1608  | 1163  | 794   | 255   | 119   | 286   | 627   | 1149  | 1895  | 2211  |
|                                                                               | CDD50       | 950    | 0     | 0     | 0     | 0     | 26    | 268   | 413   | 208   | 35    | 0     | 0     | 0     |
|                                                                               | CDD65       | 116    | 0     | 0     | 0     | 0     | 1     | 37    | 66    | 12    | 0     | 0     | 0     | 0     |
|                                                                               | CDH74       | 968    | 0     | 0     | 0     | 0     | 11    | 352   | 527   | 78    | 0     | 0     | 0     | 0     |
|                                                                               | CDH80       | 206    | 0     | 0     | 0     | 0     | 3     | 78    | 112   | 13    | 0     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 5.2    | 4.7   | 4.8   | 5.5   | 5.8   | 5.8   | 5.0   | 4.6   | 4.9   | 5.2   | 5.5   | 5.0   | 5.0   |
| Precipitation                                                                 | PrecAvg     | 22.1   | 1.4   | 0.9   | 1.0   | 1.3   | 1.7   | 2.0   | 2.3   | 2.8   | 2.4   | 2.6   | 2.0   | 1.7   |
|                                                                               | PrecMax     | 29.8   | 2.5   | 2.0   | 2.0   | 3.1   | 3.8   | 3.5   | 7.2   | 5.0   | 4.6   | 5.5   | 3.3   | 3.1   |
|                                                                               | PrecMin     | 14.7   | 0.2   | 0.2   | 0.1   | 0.3   | 0.6   | 0.6   | 0.3   | 0.6   | 0.6   | 1.0   | 0.8   | 0.7   |
|                                                                               | PrecStd     | 3.8    | 0.6   | 0.4   | 0.5   | 0.6   | 0.7   | 0.7   | 1.4   | 1.1   | 1.0   | 1.0   | 0.6   | 0.6   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 23.7  | 29.4  | 43.8  | 56.9  | 73.7  | 86.4  | 87.0  | 81.3  | 70.9  | 52.7  | 35.6  | 29.9  |
|                                                                               |             | MCWB   | 23.3  | 28.2  | 35.0  | 44.4  | 56.9  | 65.9  | 69.3  | 67.0  | 59.4  | 48.3  | 34.8  | 29.2  |
|                                                                               | 2%          | DB     | 16.8  | 24.8  | 37.8  | 51.1  | 65.9  | 82.3  | 83.1  | 75.3  | 64.2  | 47.4  | 33.1  | 27.0  |
|                                                                               |             | MCWB   | 16.0  | 23.5  | 32.2  | 40.8  | 51.8  | 63.9  | 66.7  | 64.4  | 55.8  | 43.9  | 32.0  | 26.1  |
|                                                                               | 5%          | DB     | 12.7  | 19.9  | 33.9  | 46.2  | 60.2  | 78.2  | 79.9  | 71.1  | 59.5  | 43.9  | 31.0  | 23.1  |
|                                                                               |             | MCWB   | 11.8  | 18.8  | 30.3  | 38.0  | 49.6  | 62.3  | 65.5  | 61.8  | 53.0  | 41.3  | 30.0  | 22.2  |
|                                                                               | 10%         | DB     | 8.8   | 16.5  | 31.1  | 42.0  | 54.8  | 73.9  | 76.4  | 67.3  | 55.7  | 40.7  | 27.9  | 18.9  |
|                                                                               |             | MCWB   | 7.9   | 15.5  | 28.2  | 35.8  | 45.9  | 60.7  | 64.2  | 60.0  | 51.2  | 38.4  | 26.8  | 18.1  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.4  | 28.6  | 36.0  | 45.0  | 58.5  | 68.2  | 72.4  | 69.8  | 60.7  | 49.3  | 35.0  | 29.0  |
|                                                                               |             | MCDB   | 23.8  | 28.9  | 40.8  | 55.5  | 70.6  | 80.2  | 82.7  | 78.8  | 68.1  | 52.4  | 35.6  | 29.5  |
|                                                                               | 2%          | WB     | 16.0  | 23.7  | 33.1  | 41.5  | 53.8  | 66.2  | 69.1  | 65.6  | 57.0  | 44.4  | 32.5  | 26.0  |
|                                                                               |             | MCDB   | 16.5  | 24.7  | 36.7  | 49.3  | 63.1  | 77.6  | 79.1  | 72.1  | 62.5  | 46.3  | 33.1  | 26.8  |
|                                                                               | 5%          | WB     | 11.9  | 19.0  | 30.7  | 38.9  | 50.1  | 64.4  | 67.1  | 63.2  | 54.2  | 41.8  | 30.0  | 22.2  |
|                                                                               |             | MCDB   | 12.6  | 19.9  | 33.7  | 44.8  | 59.3  | 75.5  | 76.7  | 69.4  | 58.3  | 43.8  | 31.0  | 22.9  |
|                                                                               | 10%         | WB     | 8.0   | 15.4  | 28.2  | 36.4  | 46.4  | 62.2  | 65.4  | 60.9  | 51.3  | 38.7  | 27.0  | 18.0  |
|                                                                               |             | MCDB   | 8.7   | 16.4  | 31.0  | 41.5  | 53.6  | 71.9  | 74.0  | 66.4  | 55.2  | 40.7  | 28.1  | 18.7  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 13.4  | 14.6  | 17.4  | 16.7  | 14.8  | 17.2  | 16.1  | 13.6  | 11.4  | 8.6   | 12.6  | 12.3  |
|                                                                               |             | MCDBR  | 14.3  | 14.7  | 14.9  | 18.9  | 22.8  | 23.7  | 22.4  | 20.3  | 17.9  | 10.0  | 10.2  | 12.6  |
|                                                                               |             | MCWBR  | 14.1  | 14.4  | 11.7  | 12.3  | 13.4  | 11.2  | 11.0  | 11.3  | 11.0  | 7.7   | 10.3  | 12.5  |
|                                                                               | 5% WB       | MCDBR  | 14.0  | 15.0  | 13.9  | 15.9  | 20.9  | 19.6  | 19.4  | 17.2  | 14.3  | 9.3   | 10.6  | 12.8  |
|                                                                               |             | MCWBR  | 13.9  | 14.6  | 11.8  | 11.3  | 13.0  | 10.6  | 10.6  | 10.4  | 10.0  | 7.8   | 11.0  | 12.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.244  | 0.289 | 0.330 | 0.354 | 0.369 | 0.387 | 0.439 | 0.412 | 0.349 | 0.306 | 0.231 | 0.229 |       |
|                                                                               | taud        | 1.900  | 1.993 | 1.988 | 2.043 | 2.236 | 2.374 | 2.200 | 2.252 | 2.403 | 2.226 | 1.938 | 1.869 |       |
|                                                                               | Ebn at Noon | 105    | 193   | 233   | 258   | 267   | 265   | 243   | 239   | 235   | 191   | 109   | 47    |       |
|                                                                               | Edh at Noon | 11     | 23    | 36    | 42    | 38    | 34    | 39    | 34    | 24    | 19    | 10    | 6     |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 77     | 334   | 816   | 1374  | 1643  | 1779  | 1566  | 1062  | 625   | 249   | 104   | 33    |       |
|                                                                               | RadStd      | 9      | 27    | 62    | 148   | 176   | 154   | 168   | 134   | 87    | 42    | 19    | 3     |       |

## Historical Trends

|                        |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only           |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
|                        |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (3 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page